When to use Flink SQL `LATERAL VIEW` vs. a `ProcessTableFunction` (PTF)

- 1.0 The guideline
- 2.0 A practical version of the rule
- 3.0 What "minimizing the complexity of the UDF" looks like in practice

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1.0 The guideline

Use SQL for everything it can express. Reach for a PTF only when you need something SQL fundamentally cannot do — and when you do, make the PTF do *only that thing*.

Why this is the right default:

1. **The optimizer sees SQL; it cannot see inside a UDF.** Predicate pushdown, projection pruning, join reordering, filter fusion, watermark alignment — all stop at the boundary of a Java function. Every column referenced in `eval()` must be materialized whether you use it or not.

2. **SQL is declarative and portable.** A `LATERAL VIEW` query runs unchanged across Flink versions, Confluent Cloud, and local test environments. A PTF is compiled Java—packaged, registered, versioned, and deployed.

3. **State and timers are where complexity lives.** That's where bugs hide: TTL, key cardinality, checkpoint size, late events, timer storms. Keeping that complexity isolated in a small, well-defined PTF is exactly what you want.

4. **Schema evolution is cheaper in SQL.** Adding a column to an `UNNEST` chain is trivial. Adding it to a PTF means updating type hints, output construction, tests, and rebuilding the JAR.

2.0 A practical version of the rule

Think of this as a decision ladder—stop at the first rung that works:

1. **Pure SQL** (projections, filters, joins, window TVFs, `MATCH_RECOGNIZE`)

2. **Flink SQL + built-in table functions** (`UNNEST`, `LATERAL`, `JSON_TABLE`, `STRING_SPLIT`, …)

3. **Scalar / table / aggregate UDFs** Small custom logic with no cross-row memory

4. **Row-semantic PTF** When per-row logic is too complex for a scalar UDF but still stateless *(In practice, this is often still better expressed in SQL.)*

5. **Set-semantic PTF (state + timers)** When you need keyed state, event-time timers, custom windowing, sessionization, deduplication with memory, or "emit on Nth event" 👉 **This is where PTFs justify their existence**

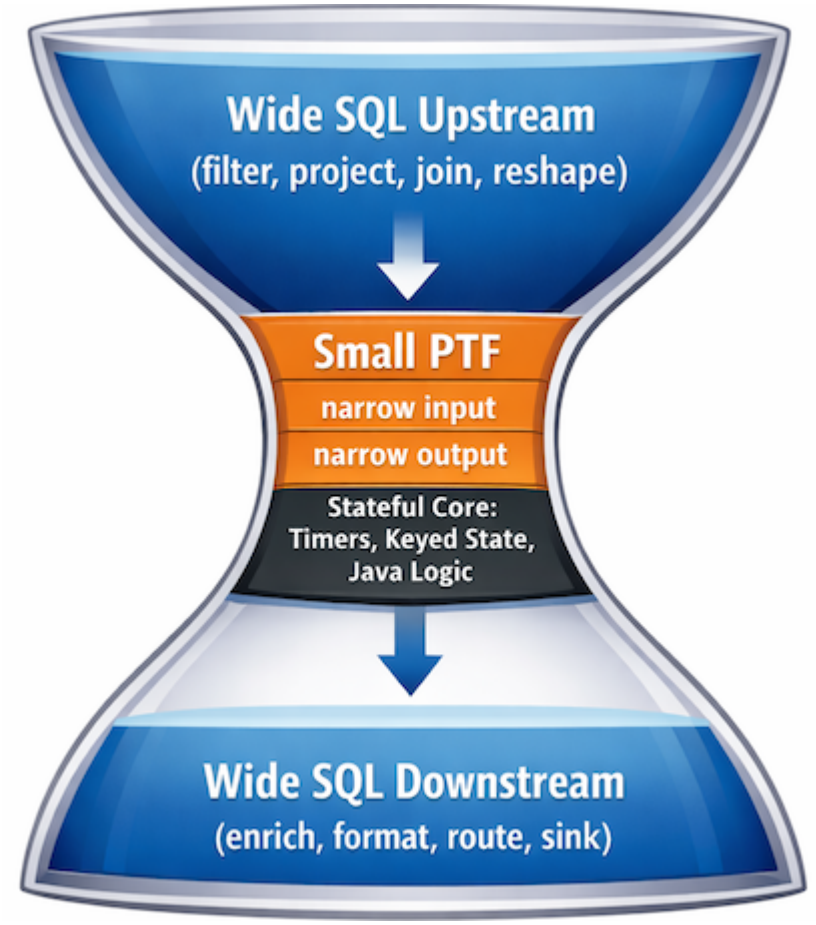
3.0 What "minimizing the complexity of the UDF" looks like in practice

When you *do* write a stateful PTF:

- **Push everything upstream into SQL.** Filter, project, and join *before* the PTF. Feed it the narrowest possible table. This reduces state size, checkpoint cost, and CPU.
- **Push everything downstream into SQL.** Emit only the minimal facts. Let SQL handle enrichment, formatting, and final shaping.
- **One PTF, one responsibility.** No “god PTFs.” If it sessionizes *and* dedupes *and* enriches — you’ve gone too far.
- **Keep `eval()` boring.** If you’re doing parsing or heavy transformations, it likely belongs in SQL or a small stateless UDF.
- **Make state explicit and named.** This pays off during debugging and checkpoint recovery.
- **Test in isolation.** A well-designed PTF has a narrow contract — table in, table out — which makes it easy to validate with a mini-cluster.

🧠 Final Mental Model

A good PTF in a SQL pipeline is shaped like an **hourglass**:



The PTF is the pinch point. Keep everything declarative in the wide top and bottom. The Java in the middle stays small and focused, used only where **process time/event time, keyed state, and control over when**

to emit, what to emit, and why go beyond Flink SQL.

💡 **If your PTF is doing formatting, joining, or reshaping — you wrote it too big.**